

Always Winning Plinko: Vision-Guided Autonomous Plinko System

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Class: MEEN 433/667 Fall 2025



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1. Objective and Significance

1.1. Project Goal

The objective of this project is to design and build an autonomous Plinko game featuring a vision-guided, motorized catcher. A single mobile catcher at the bottom of the exit will serve as the only winning slot. The system will use a front-mounted camera to track a falling ball in real-time, controlling the catcher's linear position to intercept it and create an "always-win" experience. The core of the project is the seamless integration of computer vision, real-time control systems, and electromechanical actuation to solve a time-critical interception problem. The primary goal is to demonstrate a system in which vision-based sensing enables dynamic actuation, requiring the catcher to accurately position itself before the ball arrives, with no opportunity for correction after a miss.

1.2. Technical Significance

This project addresses several critical engineering challenges that are representative of real-world robotic applications.

- **Latency-Constrained Interception:** The system must make time-critical decisions. Vision processing introduces inherent delays, which requires the control system to act on outdated information without a second chance to catch the ball. A response delay of even 50 milliseconds could result in a failure, making optimization of the entire processing pipeline essential.
- **Trajectory Uncertainty:** Collisions between the ball and the pegs introduce significant randomness, making the ball's path unpredictable from its initial conditions alone. This demands a robust, real-time tracking and prediction system rather than a simple pre-calculated trajectory. A stationary catcher would fail, as the ball's exit position varies with each drop.
- **Dynamic Actuation:** The motor must accelerate the catcher from a resting state to the correct position within a very tight timing window, demanding a control system that balances high speed with positional accuracy.

These challenges are directly applicable to industrial automation, such as high-speed pick-and-place systems, conveyor belt sorting, and robotic interception problems such as drone catching systems.

2. Design Concept

2.1. System Overview

The system consists of an open-front Plinko board viewed by a tripod-mounted camera. A Raspberry Pi 4 will perform vision processing using OpenCV to detect and track the ball. The ball's coordinates will be sent to an Arduino Uno, which will run a PID control loop to manage a motorized linear rail positioned at the board's bottom exit. The camera's

pixel coordinates will be mapped to the board’s physical coordinates using a homography transformation calibrated with markers.

2.2. Mechanical and Electrical Design

The Plinko board and pegs will be 3D printed. The catcher assembly will be a 3D-printed cup mounted on a belt-driven linear rail, actuated by a geared DC motor with an encoder for position feedback. The camera will be mounted on a tripod to maintain a fixed position for consistent calibration. Electronics will be separated into distinct power domains for the Raspberry Pi and motor systems. An L298N H-bridge will drive the motor, controlled by the Arduino. Safety is paramount; in addition to firmware-based travel limits and physical end stops, a hardware emergency stop (E-stop) button will be implemented to provide a robust physical override that cuts power to the motor.

2.3. Control and Software Concept

The control architecture uses the Raspberry Pi for high-level vision processing and the Arduino for low-level motor control. The vision software will use HSV color-based detection to identify the ball’s centroid at over 30 frames per second. These pixel coordinates are transformed into the board’s coordinate system (in mm) and sent to the Arduino via a one-way serial connection. The Arduino will use this position as the setpoint for a PID controller, which uses encoder feedback to drive the catcher motor and intercept the ball. Initially, the catcher will directly track the ball’s horizontal position; predictive algorithms may be added if system latency proves too high for this simple approach.

2.4. Representative Diagrams

The proposed system architecture and preliminary mechanical designs are shown below. It is important to note that the CAD models are conceptual and will be refined during the development phase. The dimensions provided are preliminary and may change.

The overall data and control flow is illustrated in Figure 1.

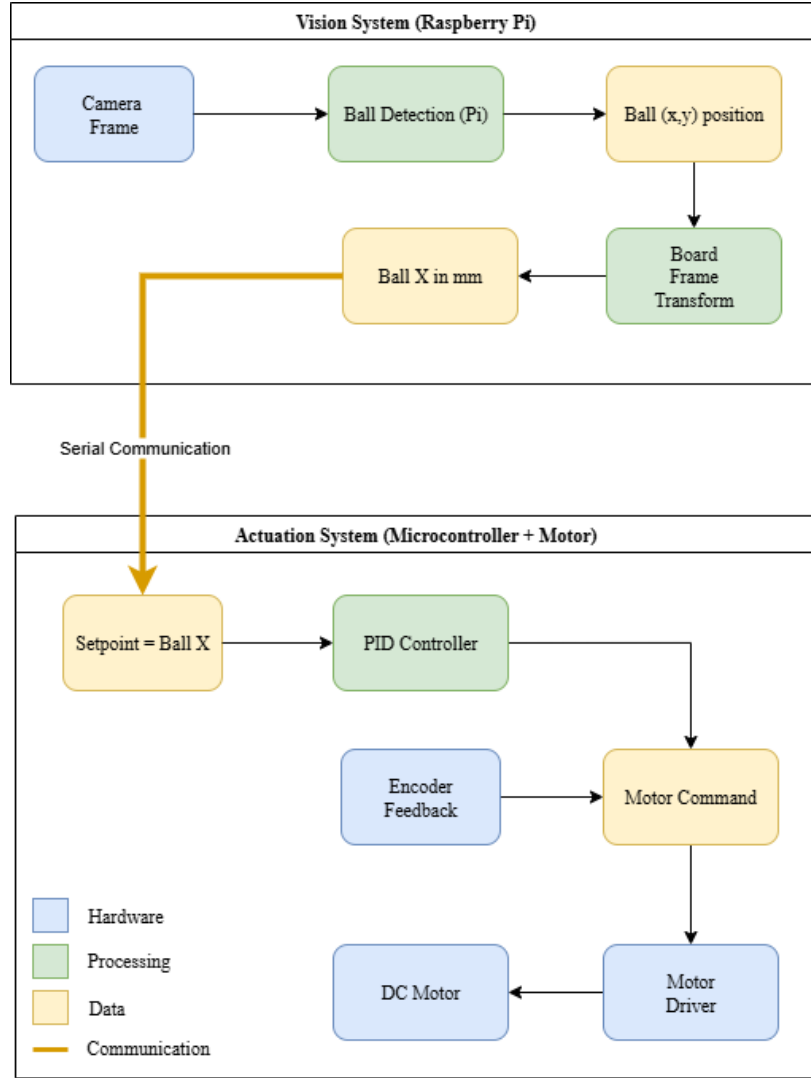


Figure 1: The system architecture, showing the data flow from vision to actuation.

A preliminary mechanical layout is provided in Figure 2, with initial dimensions detailed in Table 1.

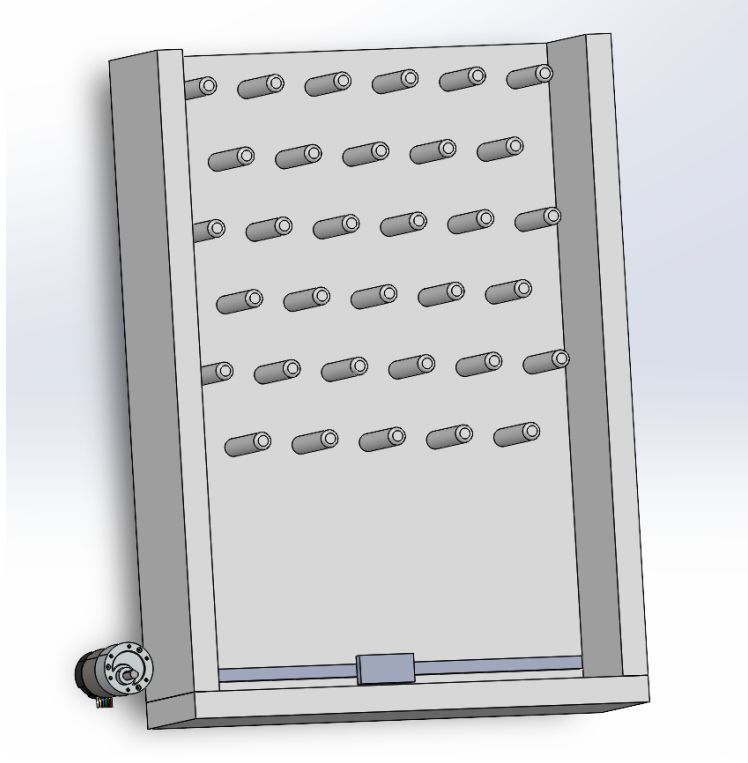


Figure 2: Preliminary mechanical concept of the Plinko board.

Table 1: Preliminary Mechanical Dimensions of the Plinko Board

Parameter	Value (mm)
Overall Board Height	457.2
Overall Board Width	304.8
Peg Field Height	266.7
Peg Field Width	266.7
Ball Diameter	25 - 40 (TBD)
Peg Diameter	12.7
Horizontal Peg Spacing	50.8
Vertical Peg Spacing	50.8

The core of the actuation system is a belt-driven linear rail. While the final design is in development, it will be based on the concept shown in Figure 3. A preliminary CAD model of the catcher is shown in Figure 4.

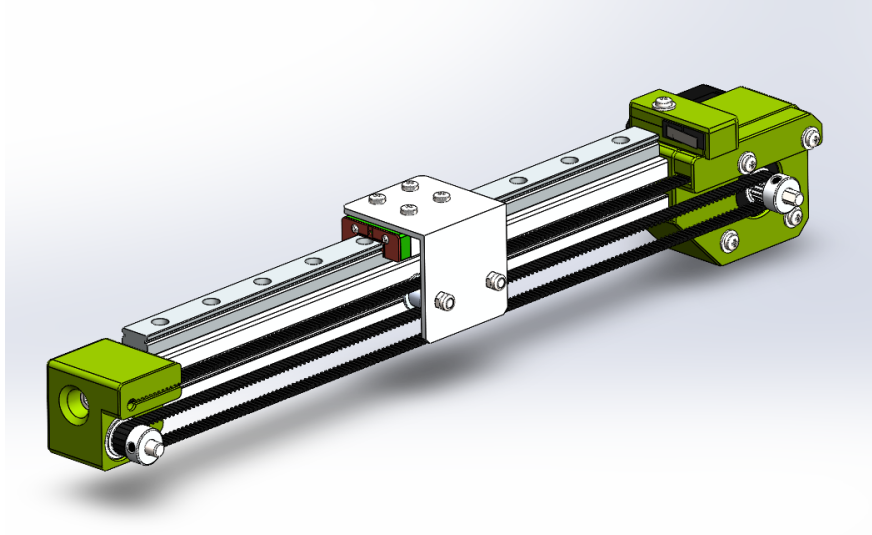


Figure 3: Reference image for the belt-driven linear rail concept.

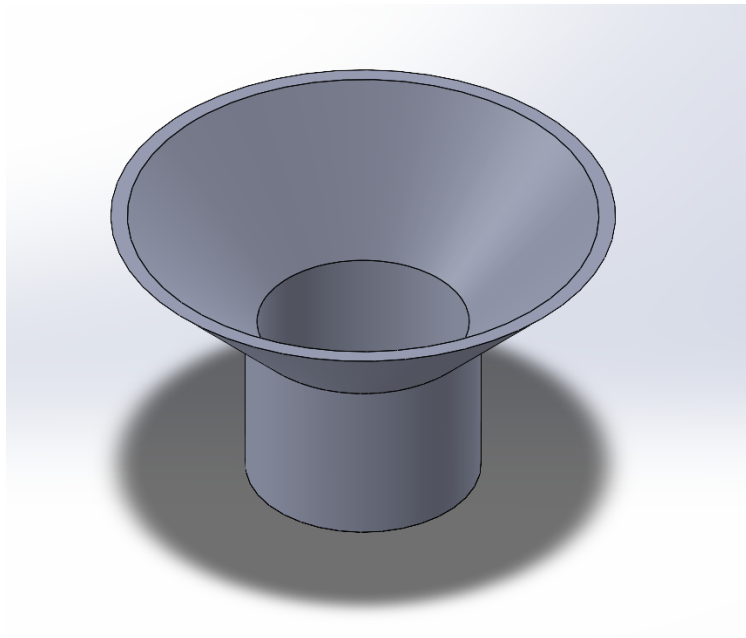


Figure 4: Preliminary 3D design of the catcher.

5. The electrical design, which includes the planned E-stop integration, is shown in Figure

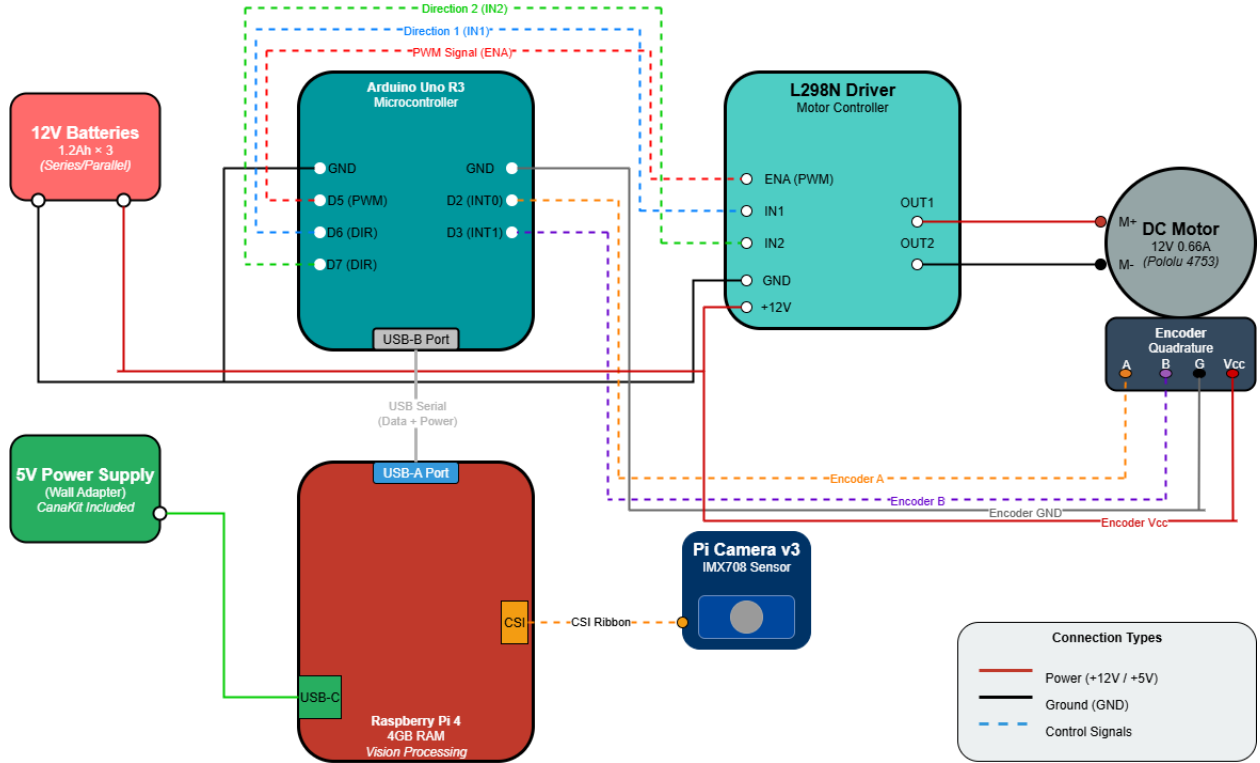


Figure 5: Circuit diagram showing power and control connections for the actuation system.

3. Work Plan

The project will be executed in three parallel workstreams followed by a final integration and testing phase.

1. Subsystem Development:

- **Mechanical:** Finalize the board and catcher design, fabricate all components using 3D printing, and assemble the linear rail, belt drive, motor, and catcher assembly.
- **Electrical:** Finalize motor and driver selection, design the power distribution and safety circuits (including E-stop), and wire all electronic components for integration testing.
- **Software:** Develop and validate the vision pipeline for reliable ball detection, implement the camera calibration routine for coordinate transformation, and establish serial communication protocols.

2. System Integration:

- Assemble the complete mechanical and electrical system.
- Establish and debug the serial communication link between the Raspberry Pi and Arduino.

- Combine the vision software and motor control firmware into a complete, end-to-end system for initial tests.

3. Testing and Refinement:

- Conduct static and dynamic tests to measure key performance metrics, including system latency and catch success rate.
- Tune the PID controller for smooth, accurate, and fast catcher motion.
- Based on baseline performance, a decision will be made on whether to implement state prediction algorithms to improve performance.

The project is scheduled to be completed by the final demonstration on December 3rd, 2025. The timeline is structured to complete subsystem development by the end of October, followed by integration in early November, leaving ample time for testing, optimization, and documentation. A detailed breakdown of all tasks, team responsibilities, and deadlines is provided in the Gantt chart located in the Appendix (see Figure 6).

4. Budget and Justification

The total budget allocated for this project is **\$500.00**. The current cost for specified components is **\$376.09**, leaving a remaining budget of **\$123.91**.

This remaining budget is critical for the completion of the mechanical systems. It will be allocated to 3D printing services for the finalized board and enclosures, as well as for purchasing any additional hardware needed for the linear rail system. This may include items such as v-slot aluminum extrusions, fasteners, and additional pulleys once the design is finalized. The budget breakdown is summarized in Table 2.

Table 2: Budget Allocation Summary

Category	Cost
Mechanical Components (Preliminary)	\$47.96
Electrical Components	\$103.42
Software & Vision Components	\$224.71
Total Purchased Items	\$376.09
Remaining for Mechanical Finalization	\$123.91
Total Project Budget	\$500.00

5. Anticipated Results

The primary deliverable will be a fully functional autonomous Plinko game that demonstrates the “always-win” concept by successfully intercepting the falling ball. The project’s success will be evaluated based on the target metrics detailed in Table 3.

Table 3: Target Performance Metrics

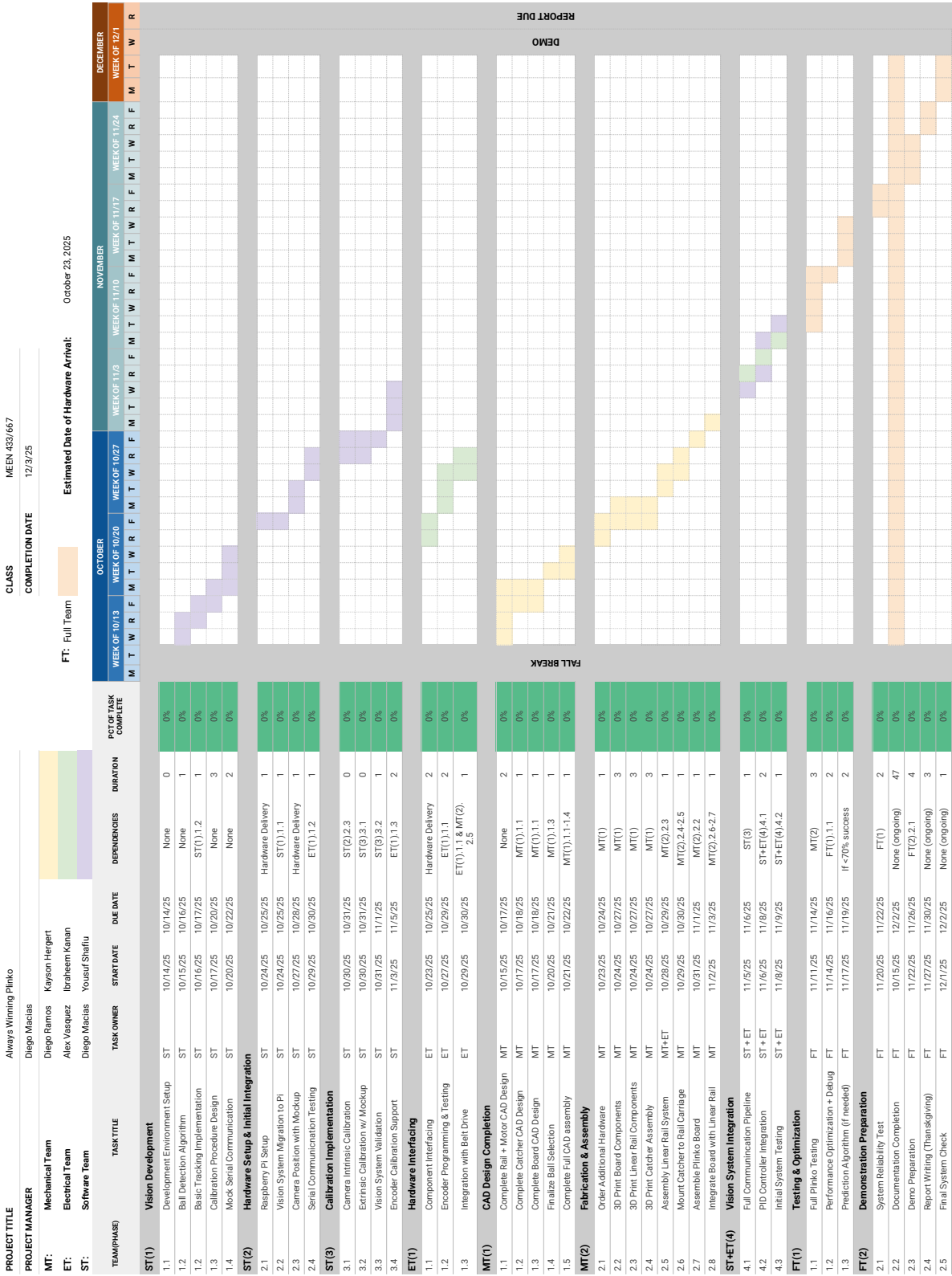
Metric	Target	Measurement Method
Ball Detection Accuracy	> 95%	Count detections vs. actual drops over 50 trials
System Latency	< 100 ms	Timestamp from image capture to motor response
Catcher Positioning Accuracy	± 5 mm	Compare commanded vs. actual encoder position
Overall Catch Success Rate	> 70%	Successful catches / total attempts over 50+ drops

The project will be considered a success if the system operates safely, achieves a catch success rate exceeding 70% over 50+ trials, integrates all subsystems seamlessly, and is completed within the \$500 budget. The project will exceed expectations if the catch rate surpasses 85% or if predictive control algorithms are successfully implemented and proven to enhance performance.

A. Project Timeline Gantt Chart

The complete project timeline is detailed in the Gantt chart below (Figure 6). This chart outlines all major tasks, dependencies, and deadlines for the mechanical, electrical, and software teams.

Always Winning Plinko Timeline



FALL BREAK

REPORT DUE

Figure 6: Detailed project timeline and Gantt chart.